San Diego Crime Trends & Spatial Hotspot Analysis (2022–2025)

Applied Empirical Research Paper & Policy Recommendations

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Repository: San_Diego_Crime_Analysis
Dataset: SDPD Police NIBRS Offenses *(N = 319,838)*

Executive Summary

I used 319,838 incident records from the San Diego Police Department (SDPD), covering 2022 through 2025, to look at how reported crime has changed across San Diego. I wanted to go beyond the annual totals and get a better sense of what was happening within them: which offenses were changing, where incidents were concentrated, and whether certain days or time periods stood out. I approached the project through exploratory data analysis, geographic concentration analysis, and statistical testing.

The overall number of reported incidents stayed relatively close to 80,000 per year. Looking at that number alone, it would be easy to assume that crime patterns were not changing very much. The breakdown by offense category and neighborhood showed otherwise. Shoplifting increased by 257.49%, the 13 highest-volume neighborhoods accounted for 38.43% of incidents with recognized neighborhood labels, and the temporal analysis identified Friday as the highest-volume day when paired with the time-shift categories. These findings show why the overall incident count does not tell the full story when considering how public safety resources are allocated (Cohen and Felson 589).

Introduction & Research Framework

Public safety leaders have to make decisions about where officers are needed, how shifts should be scheduled, and how limited resources should be distributed. I approached this project with those decisions in mind. My background in criminology gave me a foundation for understanding offense categories and the theories behind crime patterns. I plan to build on that by applying the analytical methods I have been learning in business analytics and using the data to identify which patterns the numbers support.

I organized the research around six questions. Each one gave me a different way to examine the data and connect the results to established criminological and analytical theories:

1. Macro Volumetric Trajectory (Policy Impact Theory): How has overall reported crime shifted Year-over-Year (YoY) across San Diego between 2022 and 2025? I started with the citywide totals to establish a baseline. This made it possible to see whether changes in individual offense categories were happening alongside a broader increase or decrease in reported incidents.
2. Category Dynamics (Uniform Crime Reporting / NIBRS): Which specific offense categories are driving changes in total volume, and which are growing the fastest? Looking at the NIBRS categories helps identify what is behind the overall numbers,

including increases in some offenses that declines in others might otherwise hide (Federal Bureau of Investigation).

3. Geospatial Concentration (Criminological Hotspot Theory): Is crime evenly spread across the city or concentrated in specific neighborhoods and police beats? I wanted to determine whether incidents were distributed across San Diego or clustered in a smaller number of areas. That distinction matters when considering whether targeted patrols could be more effective than evenly distributing resources (Sherman et al. 28; Weisburd 133).

4. Temporal Convergence (Routine Activity Theory): What days and operational approval-time periods have the highest number of records? Routine Activity Theory examines how motivated offenders and suitable targets converge in time and space. I used this question to identify patterns that could inform patrol scheduling (Cohen and Felson 589).

5. Severity Shifts (Severity & Offense Classification): How do trends in violent crime compare with property crime over time? Separating these categories helps show whether changes are occurring in offenses involving direct threats to personal safety or in offenses involving property loss (Federal Bureau of Investigation).

6. Seasonal Patterns (Temporal Seasonality): What recurring monthly fluctuations appear across the four calendar years? I want to determine whether certain months consistently have higher incident counts and whether those patterns can help agencies plan staffing ahead of busier periods.

Data Preparation & Feature Engineering

The primary dataset contains 319,838 incident records from January 1, 2022, through December 31, 2025, with 30 original attributes (City of San Diego Open Data Portal). Before running the analysis, I combined the four annual datasets. I reviewed the fields to make the records more consistent and to create variables that would allow me to compare incidents across categories, locations, and time periods. The main preparation steps were:

- Data Hygiene and Text Standardization: I standardized several text attributes using Title Case and trimmed extra spaces. For the geographic concentration analysis, I kept neighborhood records separate from entries labeled *Unknown / Unspecified*, *Unknown*, or *None*. Those unspecified entries were excluded from the neighborhood counts rather than treated as an actual geographic area.

- Operational Shift Binning: I grouped hours into four six-hour periods: Night (00:00–05:59), Morning (06:00–11:59), Afternoon (12:00–17:59), and Evening (18:00–23:59). One distinction in the notebook is important here: I derived the day of the week from the incident's occurred_on field, but derived the hour and time-shift category from approved_on. The shift analysis therefore reflects the time the record was

approved, not necessarily the time the incident occurred. I account for that limitation when interpreting the temporal results.

- Severity Feature Synthesis: I created a severity variable using the dataset's existing violent_crime and property_crime indicators. Records flagged as violent were classified as Violent Crime (7.6% of incidents); records flagged as property crime were classified as Property Crime (28.0%); and the remaining records were grouped as Society/Other (64.5%). I used these groupings to compare broader patterns while keeping the original offense categories available for the more detailed analysis (Federal Bureau of Investigation).

- Reviewing "All Other Offenses": "All Other Offenses" was the largest category in the top-ten offense summary, with 65,897 incidents across the four years (16,640 in 2022, 15,860 in 2023, 16,717 in 2024, and 16,680 in 2025). I kept this category in the analysis because it is part of the source data, but I would be careful about concluding it on its own. The notebook does not distinguish among the underlying offenses in this catch-all category.

Statistical Hypothesis Testing & Results

After preparing the data, I used four formal hypothesis tests and a separate Pareto concentration analysis. I wanted to check whether the patterns I noticed in the descriptive analysis were supported by statistical testing. Each test addressed a different relationship in the data, so I interpreted the results based on what was actually included in each comparison:

- Test 1: Offense Type Shift Over Time (Chi-Square Test of Independence)
    - *Question:* Did the distribution of the ten most frequently reported offense categories change between 2022 and 2025?
    - *Result:* χ^2=4,307.3914, dof=27, p<0.001.
    - *Verdict:* Reject H0. The distribution of the ten most frequently reported offense categories differed significantly by year. The notebook output rounds the p-value to 0.0000e+00, so I report it as p<0.001 rather than as zero. This is consistent with the increase in shoplifting and declines in vehicle-related theft. The test was run on the top ten offense categories overall, rather than every offense category in the dataset, so the result applies to that group.

- Test 2: Crime Severity vs. Time Shift (Chi-Square Test of Independence)
    - *Question:* Is the distribution of crime severity associated with the time-shift category?
    - *Result:* χ^2=7,402.7796, dof=6, p<0.001.
    - *Verdict:* Reject H0. The distribution of severity categories differed significantly across the four time-shift categories. The test establishes an association between severity classification and the time-shift variable, but it does not show that one factor causes the other. The time-shift variable was also created from approved_on, not the incident occurrence timestamp. The result therefore shows

an association with record approval time; it should not be interpreted as proof that those offenses occurred during those shifts.

- Test 3: Monthly Seasonality Analysis (Kruskal-Wallis H-Test)
    - *Question:* Do incident counts differ across calendar months?
    - *Result:* H=24.8189, p=0.0096896.
    - *Verdict:* Reject H0. The test found statistically significant differences among the monthly incident-count groups. In the descriptive results, February had the lowest four-year total (24,765), while July had the highest (27,879). The test compares the 12 calendar-month groups using four annual incident-count observations per month (one for each year from 2022 through 2025). It supports differences across months, although the four-year period does not establish that the same pattern will continue in future years.

- Test 4: Violent vs. Property Crime Ratio Shift (Chi-Square Test of Independence)
    - *Question:* Did the ratio of violent crime to property crime change significantly between 2022 and 2025?
    - *Result:* $\chi^2=86.2251$, dof=1, $p=1.6058\times10^{-20}$.
    - *Verdict:* Reject H0. The relationship between violent and property crime changed significantly between 2022 and 2025. Property crime increased from 19,745 incidents in 2022 to 22,827 in 2025 (+15.61%), while violent crime decreased from 6,095 to 5,811 (-4.66%). This comparison helps show how the mix of reported incidents changed even as the overall citywide totals remained relatively stable.

- Test 5: Pareto Spatial Concentration Analysis (80/20 Rule)
    - *Question:* Is crime spread evenly across San Diego, or is it concentrated in a smaller number of neighborhoods?
    - *Result:* Of 136 neighborhoods with recognized labels, the 13 highest-volume neighborhoods accounted for 122,258 of 318,148 incidents (38.43%).
    - *Finding:* The distribution shows a high level of geographic concentration. The calculation uses 13 neighborhoods, or approximately 9.6% of the 136 analyzed, and excludes incidents without a recognized neighborhood label. This is a descriptive application of the Pareto principle rather than a formal hypothesis test (Sherman et al. 31; Weisburd 135).

Key Analytical Insights

Macro Dynamics & The Shoplifting Surge

The annual totals did not change dramatically: 81,409 incidents in 2022, 79,847 in 2023 (-1.92%), 80,680 in 2024 (+1.04%), and 77,902 in 2025 (-3.44%). If I had stopped at those numbers, the main takeaway would have been that reported incident volume was relatively stable. Breaking the totals down by offense category showed a different story:

- Shoplifting Surge (+257.49%): Shoplifting had the largest percentage increase among the ten most frequently reported offense categories, rising from 1,181 incidents in 2022 to 4,222 in 2025. This stood out because it moved in the opposite direction from the overall citywide trend.
- Declining Vehicle Crime: Motor vehicle theft decreased by 33.43% (from 6,453 to 4,296), while theft from motor vehicles dropped by 35.32% (from 5,855 to 3,787). Those declines help explain why the overall totals remained fairly steady despite the increase in shoplifting.

Geographic Hotspots

I used Pareto analysis to see how incident volume was distributed across San Diego neighborhoods. The results showed that incidents were not evenly distributed. The five highest-volume neighborhoods were:

1. East Village: 18,682 incidents (5.9% of all reported incidents; +17.19% 4-year growth)
2. Pacific Beach: 11,964 incidents (-8.23% 4-year reduction)
3. Midway District: 10,951 incidents (+6.78% 4-year growth)
4. Core-Columbia: 9,997 incidents (+20.74% 4-year growth)
5. San Ysidro: 9,834 incidents (-3.10% 4-year reduction)

The changes over the four years are worth looking at alongside the incident totals. Some of the highest-volume neighborhoods experienced declines, while others, including East Village and Core-Columbia, increased. That means a response based solely on historical volume could miss areas where incident counts are moving in the opposite direction. The broader Pareto result also shows that the concentration extends beyond just these five neighborhoods.

Temporal Convergence

The day-and-shift cross-tabulation showed 115,235 records in the Afternoon category, representing approximately 36.0% of the 319,832 records with a nonmissing time-shift value (six records had no usable approval timestamp). Friday afternoon had the highest count of any day-and-shift combination, with 17,783 records. I found this pattern worth examining because it identifies a specific combination with higher recorded volume. There is an important limitation, though: the weekday is based on the incident date, while the shift is based on the approval timestamp. The result should be treated as a pattern in the records, not confirmation that Friday afternoon is when the incidents themselves occurred. I would rerun this comparison using the same incident-occurrence timestamp for both variables before making patrol-scheduling decisions.

Policy Recommendations

Based on the results, I would focus attention on the areas and offense categories where the data shows the clearest changes. First, I would prioritize reviewing resource needs in high-volume neighborhoods such as the East Village, Pacific Beach, the Midway District, and the Core-Columbia. The Pareto analysis found that the 13 highest-volume neighborhoods accounted for 38.43% of incidents with recognized neighborhood labels. That concentration gives the department a reason to look more closely at targeted foot patrols, bicycle units, and community safety teams in these areas rather than assuming that every neighborhood needs the same level of additional attention (Sherman et al. 42; Weisburd 140).

Second, the increase in shoplifting deserves a more focused response. Incidents rose from 1,181 in 2022 to 4,222 in 2025, while several other major property-crime categories declined. I recommend exploring opportunities for the department to work with local merchant associations and develop loss-prevention partnerships in commercial areas with higher shoplifting volume. The results suggest that property crime should not be treated as one uniform trend.

Finally, I recommend a closer look at patrol scheduling, but I do not recommend changing shift coverage based solely on the current Friday-afternoon result. Since the notebook uses the incident date for the day of week and the approval timestamp for the time shift, that comparison needs to be rerun using a consistent occurrence timestamp first. If the same afternoon and Friday patterns remain after that adjustment, they could provide a stronger basis for reviewing shift overlaps and officer availability (Cohen and Felson 598). For me, the main takeaway is that the data gives the department a starting point for asking more specific questions about where resources may be needed. The next step is to validate the timing pattern before turning it into an operational decision.

Citations

City of San Diego Open Data Portal. Police NIBRS Offenses (2022–2025) Datasets. San Diego Police Department (SDPD), 2025, https://data.sandiego.gov/datasets/police-property-crimes/.

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Sherman, Lawrence W., Patrick R. Gartin, and Michael E. Buerger. "Hot Spots of Predatory Crime: Routine Activities and the Criminology of Place." Criminology, vol. 27, no. 1, 1989, pp. 27–56.

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